

Prisha Priyadarshini

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EDUCATION

Incoming Master's Student

M.S. in Computer Science; University Selection in Progress

Expected May 2028

Rutgers University – New Brunswick

New Brunswick, NJ

B.S. in Computer Science, Minor in Mathematics; GPA: 3.675/4.00; Dean's List (Spring 2026)

Expected May 2027

RELEVANT EXPERIENCE

AI Engineering Intern

Aug. 2026

DrRx.ai – Remote

- Developing a RAG pipeline leveraging **BM25S** retrieval and **Qwen2.5-7B-Instruct** over **2K+ pharmaceutical documents**, integrating data from PostgreSQL and AWS S3.

Machine Learning Intern

May 2026 – Aug 2026

Regeneron – Troy, NY

- Deployed a **Streamlit** application for the QA Validation and Operations Team using **TabPFN Regressor and Classifier models**, engineering **40 trend-based features** from **500K+ GMP manufacturing records** to predict equipment breakdowns, achieving a **67.8% ROC-AUC**.

Open Source Developer (DeepChem)

May 2026 – Present

Google Summer of Code – Remote

- Selected as 1 of 5 from 200+ applicants to extend DeepChem's support for large-scale molecular language models.
- Extended the **HuggingFaceModel class** by implementing the **generate() API**, implementing batched autoregressive inference, tokenizer padding, and device-aware execution.
- Developed an **OLMo wrapper class** supporting **regression, classification**, multitask learning, and causal language modeling with custom prediction heads, pretrained checkpoint loading, mixed-precision inference, **LoRA/QLoRA**, and **multi-GPU support**.

Lead Machine Learning Researcher

Jan. 2026 – Present

DIMACS, Rutgers University – New Brunswick, NJ

- Leading a study benchmarking **interpretable decision trees** against SOTA boosting models across **6 datasets and 30+ configurations**.
- Evaluating **ThresholdGuessBinarizer (TGB)** across accuracy, stability, fairness, and Rashomon sets, showing **diminishing gains from added model complexity**.

Artificial Intelligence Researcher

Sept. 2025 – June 2026

Algoverse AI – Remote

- Built a multi-agent framework studying consensus formation, running **20-round deliberation** experiments across **GPT-4.1, Mistral, LLaMA, and Phi** on subjective and objective benchmarks.
- Introduced disagreement and deference metrics surfacing hierarchical influence patterns (**up to 80% small-to-large deference**); showed prompting strategies can control multi-agent behavior.

AI Associate Developer

Oct. 2025 – Jan. 2026

Insurity (SpatialKey Team) – Remote

- Built geospatial preprocessing pipelines over **6M+ records**, integrating climate and peril data via Pandas and GeoPandas.
- Trained LightGBM peril classifiers with **SMOTE balancing** achieving a **70% accuracy and F1 score**.
- Built a **U-Net++ change-detection system** to detect before/after wildfire satellite imagery by adding shadow normalization and shadow masking to reduce false positives from terrain illumination.

Artificial Intelligence Researcher

June 2025 – Dec. 2025

Algoverse AI – Remote

- Built a hierarchical scene-captioning pipeline with dynamic stride windowing and multimodal chain-of-thought reasoning; designing BLEU-4/METEOR/CIDEr/BERTScore evaluation and frame-sampling ablations.
- Ran YouCook2 experiments on **30 RTX A6000 GPUs**, achieving **17% higher CIDEr than GPT-4o** and **14% over VLLaMA-3**.

PUBLICATIONS

- Directional Influence and Consensus Formation in Multi-Agent Systems.**
Prisha Priyadarshini, Aryan Shrivastava.
Accepted to ICML 2026 Pluralistic Alignment Workshop.
- DynaStride: Dynamic Stride Windowing with MMCOT for Instructional Multi-Scene Captioning.**
Eddison Pham, Prisha Priyadarshini, et al.
Accepted to NeurIPS 2025 7HVV Workshop (Oral) and AAAI 2026 AI4EDU Workshop.

TECHNICAL SKILLS

Languages: Python, C, C++, Java, JavaScript

AI & ML: PyTorch, Hugging Face, Transformers, scikit-learn, OpenCV, NumPy, Pandas, GeoPandas

Cloud, Systems & MLOps: AWS, Azure, Docker, CI/CD, GitHub, pytest, Jira, Anaconda, Vercel

Certifications: AWS Certified Machine Learning – Specialty; AWS Certified Cloud Practitioner